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NOTE

TOROIDAL TILINGS FROM DE BRUIJN–GOOD CYCLIC SEQUENCES

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We show how to construct an array of integers $0, \ldots, c-1$, on a torus, so that it contains all possible $m \times n$ blocks of such integers. The constructed array has size $c^m \times c^{m(n-1)}$ and contains the least possible number of elements, c^{mn}, as each distinct $m \times n$ block of integers $0, \ldots, c-1$ appears exactly once.

For the case $c=2$ and $m=n=3$, our construction produces an 8×64 array—which is a quarter the size of the array conjectured to be minimal in [2].

De Bruijn–Good cyclic sequences

A b^k-long cyclic sequence of integers $0, \ldots, b-1$ is called a de Bruijn–Good cyclic sequence if it contains all possible k-long sequences of such integers. A sequence of this type is easily obtained from an Eulerian circuit of a graph with b^{k-1} vertices and b^k directed edges. See [1] and [4] for more information on de Bruijn–Good cyclic sequences, including the number of distinct sequences.

Constructing a $c^m \times c^{m(n-1)}$ array

Our construction uses two de Bruijn–Good cyclic sequences:

(a) A c^m-long cyclic sequence of integers $0, \ldots, c-1$, containing all possible m-long sequences of such integers.

(b) A $c^{m(n-1)}$-long cyclic sequence of integers $0, \ldots, c^m-1$, containing all possible $(n-1)$-long sequences of such integers.

The (a)-sequence provides the first column of the array. The (b)-sequence's jth term gives the amount to cyclically shift column j of the array to obtain column $j+1$. The final shift of the array's last column reproduces its first column—but cyclically shifted by s places—where s is the sum of all the shifts, reduced modulo c^m.

The average value of a shift is $\frac{1}{2}(c^m-1)$, provided $n>1$, as each integer $0, \ldots, c^m-1$ occurs equally often in a non-trivial (b)-sequence. Hence

$$s = \tfrac{1}{2}(c^m-1)c^{m(n-1)} (\mathrm{mod}\, c^m),$$

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and so

if c is odd and $n \geq 2$, then $s = 0$;

if c is even and $n > 2$, then $s = 0$;

if c is even and $n = 2$, then $s = \frac{1}{2}c^m$.

In a case where c is even and $n = 2$, we say the torus has a half-twist, as the pth element in the last column is required to be adjacent (on the torus) to the qth element in the first column, for all p, q which differ by $\frac{1}{2}c^m$. (In the case $c = m = n = 2$, see [2] for a neater solution, not needing any sort of twist.)

Example: $c = 2$, $m = 2$, $n = 3$.

Using the de Bruijn–Good cyclic sequences 0011 and 0010203112132233 produces:

	0	0	0	0	0	1	1	1	1	0	1	1	1	0	1	0
A 4×16 (toroidal) array	0	0	0	1	1	0	0	1	0	0	1	0	1	0	1	1
containing all 2×3 blocks:	1	1	1	1	1	0	0	0	0	1	0	0	0	1	0	1
	1	1	1	0	0	1	1	0	1	1	0	1	0	1	0	0

Cyclic (upward) shifts:	0	0	1	0	2	0	3	1	1	2	1	3	2	2	3	3

Generalization

Our construction generalizes to d-dimensional arrays, using d de Bruijn–Good cyclic sequences, and shifting cyclically in up to d-1 dimensions. E.g. it can be used to construct a $4 \times 4 \times 16$ array containing all $2 \times 2 \times 2$ cubes of '0's and '1's.

Addendum

After the first version of this note had been submitted, Dr C.R.J. Clapham kindly sent me [3]. It describes two methods of constructing minimal binary arrays (our case $c = 2$) and exhibits an 8×8 array for the case $c = m = 2$ and $n = 3$ of our example. Generalizing a method of that paper (to allow $c \geq 2$) leads to our construction method.

References

[1] C. Berge, The theory of graphs and its applications (Methuen, London, 1962).

[2] C.R.J. Clapham, Universal tilings and universal (0, 1)-matrices, Discrete Math. 58 (1986) 87–92.

[3] C.T. Fan, S.M. Fan, S.L. Ma and M.K. Siu, On de Bruijn arrays, Ars Combin. 19A (1985), 205–213.

[4] M. Hall, Jr., Combinatorial theory, second edition (Wiley, New York, 1986).